**Course Code:21CS54**

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# R.V.COLLEGE OF ENGINEERING

**(Autonomous Institution Affiliated to VTU)**

**V Semester B.E**

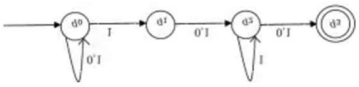
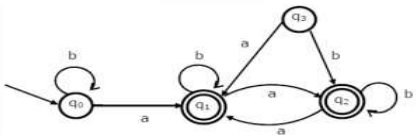
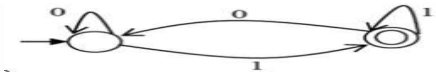
**Model Question paper**

# THEORY OF COMPUTATION

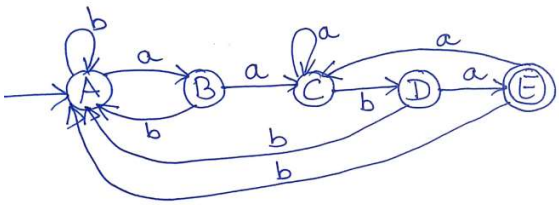
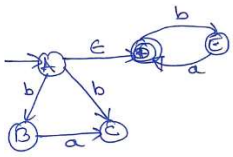
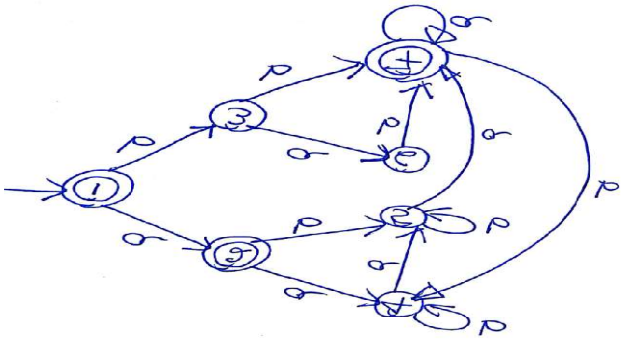
***Time: 03 Hours*** ***Maximum Marks: 100***

***Instructions to Candidate:***

1. **Answer all questions from PART-A. PART-A questions to be answered in first three pages of the answer book.**
2. **Answer questions 2, 7, 8 and any one question from 3 and 4, and any one question from 5 and 6 from PART-B.**

| | **PART-A** | |
|---|---|---|
| **1.1** | Consider the following automata. What is the set of reachable states for the input string 0011?  | **1** |
| **1.2** | The minimum state automaton equivalent to the below FA has how many number of states  | **1** |
| **1.3** | Find $R^2_{12}$=?  | **1** |
| **1.4** | Define ε –closure (q), where q ε Q of an automata. | **1** |
| **1.5** | In each case, find a string of minimum length in ∑= {0, 1} not in the language corresponding to the given regular expression.<br>i. $(0^*+1^*)(0^*+1^*)(0^*+1^*)$<br>ii. $0^*(100^*)^*1^*$ | **1** |
| **1.6** | Describe the decision algorithm to answer the following question. | **1** |

| | | |
|---|---|---|
| | Given two DFA's M1 and M2, are there any strings that are accepted by neither? | |
| **1.7** | Give CFG for the language *"The set of odd length strings in {a, b}* with the middle symbol a"*. | **1** |
| **1.8** | Show that the grammar with indicated productions is ambiguous.<br>S→aSb \| aaSb \| ϵ | **1** |
| **1.9** | Define Left Most Derivation (LMD).<br>Give LMD for abaaba in the grammar with productions<br>S→aSaSbS \| aSbSaS \|bSaSaS \| ϵ | **1** |
| **1.10** | Identify the nullable variables in the grammar given below.<br>S→AB\|ABC, A→BA\|BC\|a\| ϵ, B→AC\|CB\|b\|ϵ, C→AB\|BC\|c | **1** |
| **1.11** | If G is a context free grammar in CNF and x ϵ L(G) with \|x\|=n. How many steps are there in the derivation of x in G? | **1** |
| **1.12** | Let G be the CFG with productions S→S+S\|S-S\|S*S\|S/S\|(S)\|a<br>How many derivation strings are there for the string a+(a*a)/a-a? | **1** |
| **1.13** | DFA whose transition table is as below, here state A is the start state and states B, C are final states.<br>δ: a, b<br>A: B, B<br>B: C, C<br>C: D, D<br>D: A, B<br>What is the equivalent left linear grammar for the above DFA. | **1** |
| **1.14** | Consider the right linear grammar with the productions<br>A→aB \| bD, B→aB \| bC \| €, C→aB \| bC \| €, D→aD \| bD<br>Find the left linear grammar which is equivalent to the above right linear grammar. | **1** |
| **1.15** | Define Deterministic Push Down Automata. | **1** |
| **1.16** | What is the language generated by the Context Sensitive Grammar with the productions S→ABCS \| ABC, AB→BA, AC→CA, BC→CB, BA→AB, CA→AC, CB→BC, A→a, B→b, C→c. | **1** |
| **1.17** | Obtain Turing Machine to accepts the language<br>L={w\| w ϵ {a, b}* and w ends with abb}. | **1** |
| **1.18** | What is the language generated by the right linear grammar with productions<br>S→aA \| bC \| b, B→aC \| bA \| a, A→aS \| bB, C→aB \| bS. | **1** |
| **1.19** | If L1 and L2 are recursively enumerable languages, then L1⋂ L2 is recursively enumerable language. Is this statement is TRUE or FALE. | **1** |
| **1.20** | What is the solution to the instance of PCP given below.<br>List A: Xi; List B: Yi<br>1: 1, 010<br>2: 0, 10<br>3: 10, 101<br>4: 01, 100 | **1** |

| | PART-B | |
|---|---|---|
| **2 a** | Find the regular expressions corresponding to each of the following subsets of {0,1}*.<br>a) The language of all strings containing exactly two 0's.<br>b) The language of all strings that do not end with 01. | **4** |
| **b** | For the DFA below describe, either in words or by writing regular expressions, the strings that cause the DFA to be in each state.<br>  <br> | **4** |
| **c** | Explain the algorithm to find an equivalent NFA from the given NFA- ϵ. Use this algorithm to draw an NFA for the NFA- ϵ given below.<br>  <br> | **4** |
| **d** | For the DFA shown below, use the minimization algorithm to find a minimum state DFA recognizing the same language.<br>  <br> | **4** |
| **3 a** | State and prove pumping lemma for regular languages. Use pumping lemma to show that the following languages are not regular.<br>a) L={ww \| w ϵ{0,1}*}.<br>b) L={xy \| x,y ϵ{0,1}* and y is either x or $x^r$}. | **6** |
| **b** | Describe decision algorithm to answer the following questions.<br>a) Given two DFA's M1 and M2, are there any strings that are accepted by neither.<br>b) Given a regular expression r and an DFA M, are the corresponding languages are same? | **4** |

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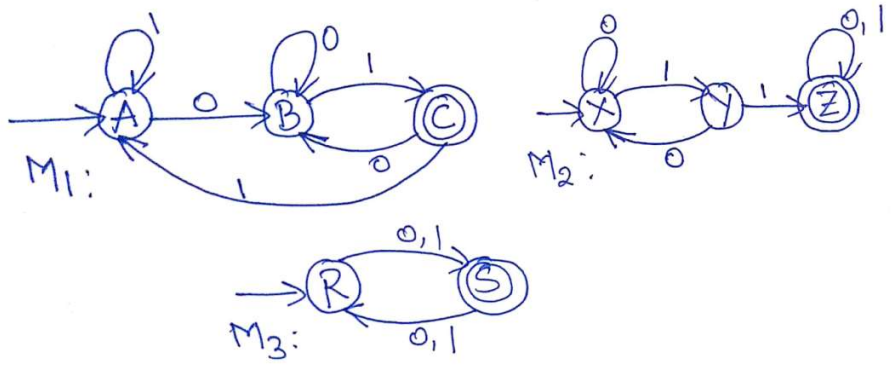
|---|---|---|
| **c** | Let M1, M2 and M3 are the DFA's pictured below recognizing languages L1, L2 and L3 respectively.<br>  <br><br>Draw DFA's recognizing the following languages.<br>a) L1 U L2<br>b) L1∩ L2<br>c) L1-L2 | **6** |
| | **OR** | |
| **4 a** | Define CFG, Language generated by CFG, Left most derivation and Right most derivation. Consider the grammar below S→S+S \| S-S \| S*S \| S/S \| (S) \| a.<br>i. Give two left most derivations for the string a+(a*a)/a-a.<br>ii. Give the derivation tree corresponds to each of the derivations in (i).<br>iii. How many distinct left most derivations of this string are there?<br>**iv.** How many derivation trees are there for the string a+a+a+a+a? | **4** |
| **b** | In each case, given the context free grammar G, find an equivalent CFG with no useless variables.<br>i. S→ABC \| BaB, A→aA \| BaC \| aaa, B→bBb \| a, C→CA \| AC<br>ii. S→AB \| AC, A→aAb \| bAa \| a \| ϵ, B→bbA \| aaB \| AB, C→abCa \| aDb \| ϵ, D→bD \| aC | **4** |
| **c** | Define GNF grammar. Given below a CFG G, find a CFG $G^1$ in GNF generating L(G) – { ϵ }. A→BC, B→CA \| b, C→AB \| a | **4** |
| **d** | Draw the NFA accepting the language generated by the grammar with productions S→abA\|bB\|aba, A→b\|aB\|bA, B→aB\|aA | **4** |
| **5 a** | Define PDA. Construct PDA for L={$ww^R$ \| w ϵ {a,b}*}. Show by IDs the string abaaba is accepted. | **6** |
| **b** | Let L be L(M1) for some PDA with empty stack M1=(Q,∑,Γ,δ,$q_0$,$Z_0$,ϕ), prove that there exists an final state PDA M2 such that L=L(M2). Construct empty stack PDA to accept L={w \| w ϵ {a,b}* and w is palindrome}. Convert it into equivalent final state PDA. | **4** |
| **c** | What are steps to be followed while finding an equivalent PDA by empty stack for the given CFG. Find the equivalent PDA to the CFG with productions S→aSb \| bSa \| abS \| baS \| Sab \| Sba \| ϵ. Show that the string abaabb is generated by the given CFG and it is accepted by its equivalent PDA. | **6** |

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| | OR | |
| **6 a** | State and prove pumping lemma for context free languages.<br>Show that L={ww \| w ∈{a,b}*} is not context free. | **6** |
| **b** | Let L1={$a^i b^j c^k$ \| i< j } and L2={ $a^i b^j c^k$ \| i< k }. Show that L1 and L2 are context free but L3=L1∩ L2 is not context free. | **6** |
| **c** | Decide whether the following languages are CFL or not.<br>i. L={$a^n b^m a^m b^n$\| m,n≥0}.<br>ii. L={xayb\| x,yϵ{a,b}* and \|x\|=\|y\|}. | **4** |
| **7 a** | Design Turing Machine to accept the language L={ w \| ε {a, b}* and Na(w)=Nb(w)}. Trace the machine for the string *baabab*. | **6** |
| **b** | Let x and y are two positive integers represented using unary notation. Design a Turing Machine that computes the function f(x,y), where x,y ϵ $1^+$<br>f(x, y) = x - y if x > y<br>f(x, y) = y - x if y>x<br>f (x, y)=0 if x = y | **6** |
| **c** | If $L_1$ and $L_2$ are recursively enumerable languages over ∑, then show that $L_1$ U $L_2$ is also recursively enumerable languages. | **4** |
| | OR | |
| **8 a** | Design Turing Machine to perform the string reverse operation over ∑={a, b, c}. Use Instantaneous Descriptions to show operation on w=*abac.* | **8** |
| **b** | Write a note on<br>i. Multi tape Turing Machine<br>ii. Non deterministic Turing Machine | **6** |
| **c** | Prove that every regular language is also a recursively enumerable language. | **2** |
| **9 a** | Construct the Linear Bounded Automata for the language L={ $ww^R$ \| w ε {a, b}*}. Trace the machine for the string *aabbaa*. | **8** |
| **b** | Define Context Sensitive Grammar.<br>Give Context Sensitive Grammar to generate the language L={ $a^n b^n c^n d^n$ \| n ≥1}. Show that the string *aabbccdd* is generated by the grammar. | **4** |
| **c** | Write a note on Chomsky hierarchy. | **4** |
| | OR | |
| **10 a** | Define Unrestricted Grammar. Give Unrestricted Grammar to generate the language L={ w \| ε {a, b, c}* and Na(w)=Nb(w)=Nc(W)}.Show that the string *acabcb* is generated by the grammar. | **8** |
| **b** | Write a note on undecidability. | **4** |
| **c** | If the given language L is Context Sensitive Language, then L is recursively enumerable language. Prove this statement. | **4** |

**Q.P.Code**
**Course code: 21CS54** **Course Title: Theory of Computation**

**PART-A**

| Q.No | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 1.6 | 1.7 | 1.8 | 1.9 | 1.10 |
|---|---|---|---|---|---|---|---|---|---|---|
| BT | 1 | 2 | 3 | 1 | 2 | 3 | 2 | 1 | 1 | 2 |
| COs | 1 | 1 | 2 | 1 | 2 | 2 | 2 | 1 | 2 | 1 |
| | | | | | | | | | | |
| Q.No | 1.11 | 1.12 | 1.13 | 1.14 | 1.15 | 1.16 | 1.17 | 1.18 | 1.19 | 1.20 |
| BT | 3 | 4 | 4 | 4 | 1 | 3 | 3 | 2 | 1 | 4 |
| COs | 2 | 3 | 3 | 2 | 1 | 3 | 2 | 1 | 2 | 3 |

**PART-B**

| Question No | | BT level | COs Addressed | | Question No | | BT level | COs Addressed |
|---|---|---|---|---|---|---|---|---|
| 2 | a | 2 | 1 | | 3 | a | 3 | 2 |
| | b | 3 | 2 | | | b | 2 | 2 |
| | c | 4 | 3 | | | c | 4 | 3 |
| | d | 4 | 3 | | | d | | |
| 4 | a | 3 | 3 | | 5 | a | 3 | 3 |
| | b | 4 | 3 | | | b | 2 | 2 |
| | c | 4 | 4 | | | c | 2 | 3 |
| | d | 3 | 4 | | | d | | |
| 6 | a | 2 | 2 | | 7 | a | 4 | 4 |
| | b | 3 | 3 | | | b | 4 | 4 |
| | c | 3 | 4 | | | c | 3 | 2 |
| | d | | | | | d | | |
| 8 | a | 4 | 4 | | 9 | a | 4 | 4 |
| | b | 3 | 2 | | | b | 3 | 2 |
| | c | 3 | 2 | | | c | 1 | 1 |
| | d | 4 | 1 | | | d | | |
| 10 | a | 3 | 3 | | | a | | |
| | b | 2 | 1 | | | b | | |
| | c | 2 | 1 | | | c | | |
| | d | | | | | d | | |

**Signature of Scrutinizer** **Signature of Chairman**

**Name:** **Name:**